

Reflexive Spaces

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One of the questions in functional analysis is to “represent” continuous linear functionals on X , in manners such as the Riesz Representation Theorem.

One step further: can we represent continuous linear functionals on X' ?

The Evaluation Map

Let X be a normed vector space; for any $x \in X$ the “evaluation functional” $\text{ev}_x : X' \rightarrow \mathbf{R}$ which maps x' to $x'(x)$ is an element of X'' .

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$$\|x\|_X = \|Ex\|_{X''} = \|\text{ev}_x\|_{X''} .$$

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We say X is *reflexive* if E is surjective. Then X is isometrically isomorphic to X'' .

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For any $1 < p < \infty$, we have $(l_p)' = l_q$ where $pq = p + q$. Thus $l_p'' = l_q' = l_p$ so l_p is reflexive.

Properties of Reflexive Spaces

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Any closed subspace of a reflexive space is reflexive.

If X is a Banach space then X is reflexive if and only if X' is reflexive.

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$(c_0)' = l_1$ and $(l_1)' = l_\infty$ so c_0 is not reflexive. Since X is reflexive if and only if X' is reflexive, we get that l_1 and l_∞ are also not reflexive.

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$C^0([a, b], \mathbf{R})$ is also not reflexive; its dual space is the functions of bounded variation, and the dual space of $BV([a, b], \mathbf{R})$ is massive.

References



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